

About the nonlinear term

The problem we are trying to solve is written as [?]

$$i \frac{\partial}{\partial t} \phi_\omega(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \quad (1)$$

where

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \varkappa \delta(x), \quad \hat{V}_{\text{ext}} = F(t)x, \quad (2)$$

$$\phi_\omega(x, 0) = \frac{-i\varkappa^{3/2}}{\omega} \text{sgn}(x) \left(e^{-\varkappa|x|} - e^{-\varkappa(\omega)|x|} \right), \quad \psi(x, 0) = \varkappa^{1/2} e^{-\varkappa|x|}, \quad (3)$$

with

$$\hat{p} = -id/dx, \quad \text{and} \quad \varkappa(\omega) = \sqrt{\varkappa^2 + 2\omega}. \quad (4)$$

A RK4 adapted to our problem has been written and must be tested as rigourously as possible.

Let us first consider the simpler following problem

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x), \quad (5)$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \varkappa \delta(x), \quad \hat{V}_{\text{ext}} = F(t)x, \quad (6)$$

$$E_0 = -\varkappa^2/2, \quad \psi_0(x) = \varkappa^{1/2} e^{-\varkappa|x|}, \quad \varkappa = 1 \quad (7)$$

$$E_k = k^2/2, \quad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \quad R(k) = \frac{i\varkappa}{k - i\varkappa}. \quad (8)$$

The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t'), \quad 0 \leq t \leq T, \quad t' = t - T/2, \quad (9)$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad n_{oc} = 5. \quad (10)$$

Ultimately, Eq. (5) is of the form

$$\dot{y} = -iA_{n \times n} y, \quad (11)$$

where the dot denotes differentiation with respect to time and A is a square, real and hermitician matrix. Of course, given the problem, the solutions y are nonetheless complex.

1 Detailed formulation of the propagator

Following Kidd *et al.* [?], we want to write the problem (1) as

$$\frac{dy}{dt} = f(y, t), \quad y(t=0) = y_0, \quad (12)$$

$$f(y, t) = \mathcal{L}y + \mathcal{N}(y, t), \quad (13)$$

where $\mathcal{L}$ is the linear part. $\mathcal{N}(y, t)$ is a priori a nonlinear term, but we will use it as a nonhomogeneous term in the following.

Let us consider the Schrödinger equation

$$i \frac{\partial}{\partial t} \Psi(x, t) = H(x, t) \Psi(x, t), \quad (14)$$

$$= \mathcal{L}(x, t) \Psi(x, t) + \mathcal{N}(\Psi, t), \quad (15)$$

with $\mathcal{L} = H_0 + V_{\text{ext}}$, where H_0 is the unperturbed Hamiltonian and V_{ext} , is the interaction with the laser field. In the following the inhomogeneous term will be considered to assume the following form

$$\mathcal{N}(\Psi, t) = \mathcal{N}\Psi(x, t). \quad (16)$$

Using the integrating factor method, the solution will be probed under the form

$$\Phi(x, t) = e^{i\mathcal{L}t} \Psi(x, t). \quad (17)$$

Using this in the Eq. (5), we get

$$i \frac{\partial}{\partial t} (e^{-i\mathcal{L}t} \Phi(x, t)) = i \left(-i\mathcal{L}e^{-i\mathcal{L}t} \Phi(x, t) + e^{-i\mathcal{L}t} \frac{\partial}{\partial t} \Phi(x, t) \right) \quad (18)$$

$$= \mathcal{L}e^{-i\mathcal{L}t} \Phi(x, t) + ie^{-i\mathcal{L}t} \frac{\partial}{\partial t} \Phi(x, t) \quad (19)$$

Equating with the RHS, we then get

$$\frac{\partial}{\partial t} \Phi(x, t) = -ie^{-i\mathcal{L}t} \mathcal{N}(e^{-i\mathcal{L}t} \Phi, t). \quad (20)$$

Adopting the notation $f(x, t_n) = f_n$, and integrating between t_n and t_{n+1} , we get

$$\Phi_{n+1} = \Phi_n - i\Delta t e^{-i\mathcal{L}t_n} \mathcal{N}(e^{-i\mathcal{L}t_n} \Phi_n, t_n). \quad (21)$$

From Eq. (17), and setting $\Delta t = t_{n+1} - t_n$, we then get

$$\Psi_{n+1} = e^{-i\mathcal{L}\Delta t}\Psi_n - i\Delta t e^{-i\mathcal{L}\Delta t}\mathcal{N}(\Psi_n, t_n). \quad (22)$$

This last equation is the formulation of our main problem. When the inhomogeneous term is set to the null matrix, Eq. (22) reduces to the split-operator method. When the linear part is considered null, which amounts to setting $H = \mathcal{N}$, we have a full Runge-Kutta scheme. For the sake of evaluating the precision of our RK scheme, this last situation is the exact one we want to be in.

In the following, we set the initial propagation time $t_i = 0$ and the final propagation time $t_f = 2\pi$. The time step is then defined as $\Delta t = (t_f - t_i)/N$, where N is the number of propagation steps. For the sake of assessing the accuracy of our scheme, we consider the solutions obtained from the split-operator method as the exact solution. We define the relative error between the exact solution and the numerical solution (obtained from our Runge-Kutta scheme) as

$$\sigma_r(j) = \left| \frac{y(j) - y_{\text{ex}}(j)}{y_{\text{ex}}(j)} \right| \quad (23)$$

2 Test 1

3 Test 2

We now want to test our scheme by setting

$$\mathcal{L} = 0, \quad \mathcal{N} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}. \quad (24)$$

The problem is therefore the following ODE system

$$\begin{cases} y_1' = -iy_1 - 2iy_2 \\ y_2' = -i2y_1 - iy_2 \end{cases} \quad \text{with} \quad y_i^0 = y_i(t=0) = i. \quad (25)$$

We can see that the eigenvalues of the matrix are -1 and 3 , which makes the eigenvalues of this problem $r_1 = i$ and $r_2 = -3i$. The solutions can be found by hand. They read

$$y_1 = -\frac{1}{2}e^{it} + \frac{3}{2}e^{-3it} \quad \text{and} \quad y_2 = \frac{1}{2}e^{it} + \frac{3}{2}e^{-3it} \quad (26)$$

We solve this problem by twice, using both the split-operator method and Runge-Kutta and compare the results by the mean of (23). The results are presented in Tab. (1) and Fig. (??).

N	$\sigma_r(1)$	$\sigma_r(2)$
10	1.23(-2)	0.9757850346
100	8.49(-7)	2.06(-4)
1000	8.19(-11)	1.99(-8)
1000	4.44(-13)	2.26(-12)

Tab. 1: Relative error $\sigma_r(i)$ between the solutions obtained from the split operator method and RK4 scheme for the component i of the solution to the problem.

4 Test 3

Let us now consider the case

$$\mathcal{L} = 0, \quad \mathcal{N} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}. \quad (27)$$

represented by following ODE system

$$\begin{cases} y_1' = -iy_1 - iy_2 - iy_3 \\ y_2' = -iy_1 - 2iy_2 - 2iy_3, \\ y_3' = -iy_1 - 2iy_2 - 3iy_3 \end{cases} \quad \text{with} \quad y_i^0 = y_i(t=0) = i. \quad (28)$$

This problem is numerically solved using Lapack. The eigenvalues are given by $r_1 = -0.307978528369904i$, $r_2 = -0.643104132107790i$ and $r_3 = -5.04891733952230i$.

N	σ_1	σ_2	σ_3
10	3.44(-5)	1.37(-3)	242889.815911347
100	2.35(-9)	9.34(-8)	2.78(-3)
1000	2.30(-13)	9.01(-12)	2.69(-7)
10000	1.06(-12)	1.57(-14)	2.68(-11)